Business Plan

Circuit Breakers Team #17

SmartHat

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SmartHat LLC

Executive Summary

Workers in construction, mining, and industrial environments face constant exposure to hazards including falling objects, toxic gas buildup, poor visibility, and undetected medical emergencies. Traditional hard hats provide only passive impact protection and offer no real-time monitoring or alert capability. As a result, supervisors and safety personnel are often unaware of dangerous conditions until after injury or exposure has occurred.

SmartHat LLC addresses this gap by integrating active safety monitoring directly into standard personal protective equipment. SmartHat is a modular, sensor-integrated hard hat that detects impacts greater than 10G, monitors carbon monoxide levels above 50 ppm, measures UV exposure within ±1 index value, tracks heart rate within ±10 bpm, activates a headlamp in low-visibility conditions, and sends real-time cellular alerts to supervisors through LTE communication.

The estimated manufacturing cost per SmartHat unit is approximately $600. SmartHat will retail at $950 per unit, providing a gross profit margin of approximately 37%. The company projects first-year sales of 100 units ($95,000 revenue), growing to 250 units in year two and 500 units in year three as partnerships with OSHA-affiliated programs and contracting firms expand.

SmartHat LLC is positioned not only as a product developer but as a workplace safety technology company. With successful market adoption, SmartHat will expand into additional smart PPE solutions including safety vests and specialized helmets for chemical plants and emergency responders. Figure I below provides an overview of SmartHat's systems.

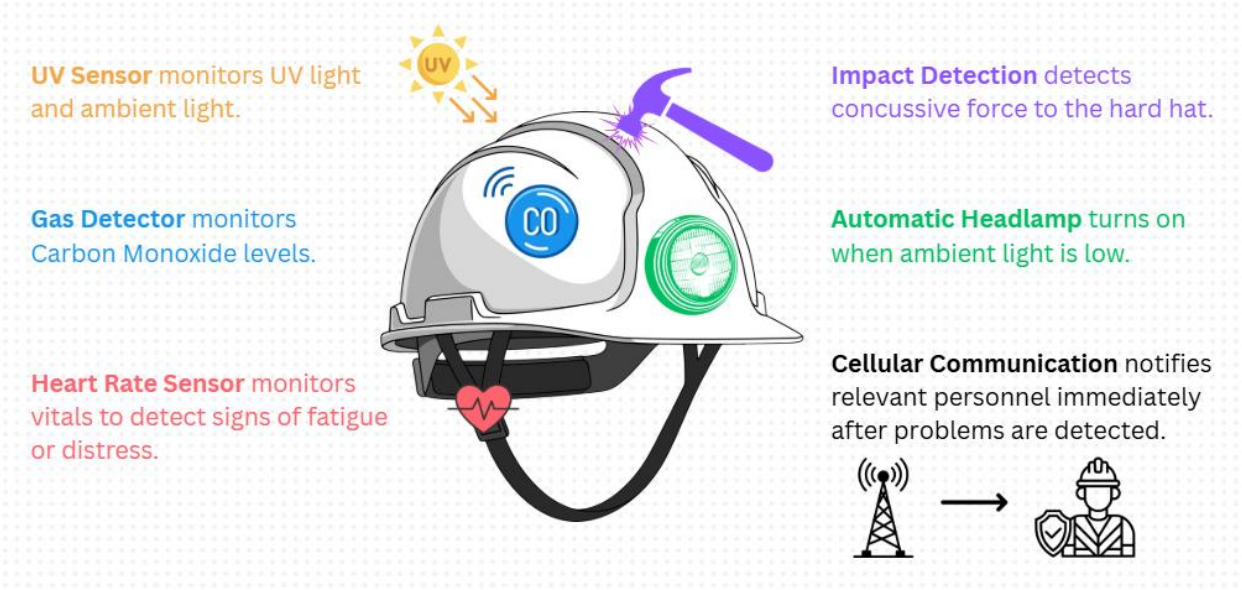

Figure I: Subsystem Overview

SmartHat uses integrated UV, gas, impact, and heart rate sensors with automatic headlamp activation and cellular alert communication.

Objectives

• Sell 100 SmartHat units in Year 1, scaling to 500 units by Year 3
• Establish partnerships with at least 20 contracting firms or OSHA-affiliated safety programs within three years
• Achieve positive net profit by the end of Year 1
• Expand product line into at least one additional PPE platform within five years

Keys to Success

• Maintain sensor accuracy and system reliability under harsh industrial conditions
• Secure credibility through OSHA alignment and contractor pilot programs
• Develop scalable assembly processes to reduce per-unit cost over time
• Maintain LTE connectivity reliability in low-coverage work environments
• Focus marketing on safety managers and compliance officers rather than individual workers

1. Plan Overview

1.1 Mission

SmartHat LLC is a workplace safety technology company dedicated to advancing active hazard detection in industrial environments. The company's core competency is integrating reliable sensing, communication, and alert systems into personal protective equipment to enhance real-time situational awareness and reduce workplace injuries.

SmartHat LLC is not simply a helmet manufacturer; it is a safety innovation company focused on embedding intelligent monitoring systems into the equipment workers already wear.

1.2 Objectives

Product-Oriented Objective:
Deploy a fully compliant, field-tested SmartHat system capable of meeting defined performance requirements including CO detection within five seconds at 50 ppm, impact detection ≥10G, track heart rate, measure UV index, and cellular alert transmission within one minute of hazard detection.

Company-Oriented Objective:
Establish SmartHat LLC as a recognized safety technology provider within regional construction and industrial markets within three years.

1.3 Keys to Success

Several key goals must be met for SmartHat to be a successful product. Firstly, the sensors must maintain reliability even under harsh industrial conditions, as SmartHat is intended to be worn in such environments. Failure to maintain integrity would compromise SmartHat's primary function, and therefore, decrease customer satisfaction. Credibility must also be secured via OSHA alignment and contractor pilot programs. This will aid in gaining the trust of the public, which will increase sales. Scalable assembly processes must be developed to reduce per-unit cost over time, which will make SmartHat competitive in the PPE market. SmartHat must also maintain LTE connectivity reliably in low coverage work environments. This will keep the products' effectiveness from being limited by the region in which it is used, thus increasing the market. Finally, the marketing for SmartHat must be targeted towards safety managers and compliance officers rather than individual workers. Individual workers may be interested in the product, but the safety managers and compliance officers will be the ones who buy the product in bulk for their company, as SmartHat is intended to be used in a system and not as an individual product.

1.4 Competitive Advantage

Existing smart helmet solutions often focus only on impact detection or fall monitoring. SmartHat differentiates itself by integrating multiple hazard-detection capabilities into a single, modular platform. These include Impact detection, Carbon Monoxide detection, Vitals monitoring, UV measurement, an automatic headlamp, and Cellular Communication alerts to safety representatives. By consolidating environmental, biometric, and communication capabilities into one architecture, SmartHat provides proactive hazard detection and rapid emergency alerts, leading to overall increased worker safety.

1.5 Target Market

SmartHat's primary target markets include construction companies and contractors, industrial manufacturing plants, infrastructure and utility contractors, and organizations running OSHA-aligned workplace safety programs. SmartHat's secondary target markets include companies making direct purchases for safety compliance upgrades, mining operations, and chemical processing facilities. Also, SmartHat will initially focus on regional markets before expanding nationally.

1.6 Basic Strategies

Startup funding will initially rely on limited capital investment and reinvested profits. Manufacturing will begin in small batches with in-house assembly to control cost and quality. Marketing efforts will focus on direct outreach to safety managers, contractor firms, and OSHA-aligned initiatives.

Future growth will include expansion into additional PPE technologies such as smart safety vests and specialized helmets for hazardous environments.

2. Company Summary

2.1 Company Description

SmartHat LLC is a Limited Liability Company founded by members of the Circuit Breakers Senior Design Team. The company develops intelligent safety-integrated personal protective equipment for industrial workers. The company will be an LLC founded by Aaron Ramsay, Dawson Carnathan, Matthew Dykes, Leigh Kergosien, and Landon Moore.

2.2 Company Location and Facilities

SmartHat LLC will initially operate from space provided by a founding member, minimizing overhead costs. Prototyping and testing may utilize university lab facilities when available.

2.3 Company Strategy

Our strategy for achieving a successful startup centers on reliability, affordability, and strong compliance alignment within high-risk work environments. SmartHat will prioritize building a dependable product that performs consistently in real-world job site conditions, ensuring contractors and safety managers can trust the technology. Early success depends heavily on demonstrating durability, long battery life, and accurate sensor performance in construction, manufacturing, and industrial settings.

To establish credibility, we plan to form partnerships with local contractors and safety-focused organizations. By aligning SmartHat with OSHA-related programs and workplace safety initiatives, we can position the product not just as wearable technology, but as a tool that supports regulatory compliance and reduces liability.

2.4 Startup Costs

Estimated Startup Expenses:
Prototype Development: $1,000
LLC Registration and Legal Fees: $150
Initial Marketing and Outreach: $1,000
Initial Component Inventory (First Batch Materials): $10,000
Insurance and Compliance Preparation: $1,000

Total Estimated Startup Expenses: ~$14,150

Startup funding will come from team contributions and reinvested revenue. Operating without rented office space reduces early fixed costs.

3. Product Summary

3.1 Product Description

The SmartHat is a sensor-integrated hard hat engineered to detect hazardous conditions in real time while maintaining full compliance with ANSI Z89.1 safety standards. It combines a suite of advanced sensors, including an electrochemical CO sensor capable of detecting concentrations of 50 ppm or higher, a triple-axis accelerometer that registers impacts of 10G or greater, and an optical PPG heart rate monitor with ±10 bpm accuracy. Environmental awareness is enhanced through a UV and ambient light sensor with ±1 index precision, while an automatic 300-lumen LED headlamp provides reliable illumination in low-light conditions. Connectivity is enabled via an LTE Cat-M1/NB-IoT module, allowing real-time cellular communication. All electronics not exposed to environmental conditions are securely enclosed to ensure durability and maintain proper balance, delivering a rugged yet comfortable safety solution for industrial applications.

3.2 Customer Needs and Benefits

Companies need real-time hazard detection to reduce injuries, liability, and downtime. Workers need early warnings and faster emergency responses. SmartHat provides immediate hazard detection, reduced response time, enhanced safety documentation, and improved safety culture.

3.3 Future Products

SmartHat LLC is exploring several exciting future product opportunities that expand beyond headwear into a broader ecosystem of smart safety solutions. These include a smart safety vest equipped with environmental sensors, as well as chemical plant–specific personal protective equipment designed for high-risk industrial settings. The lineup may also extend into emergency responder helmets built for first responders who operate in unpredictable and dangerous conditions, all tied together through integrated dashboard software that gives safety managers a centralized view of their team's safety data in real time.

3.4 Competitive Comparison

There are limited options available for smart personal protective equipment that are affordable and integrate multiple environmental safety systems into a single hard hat. Systems that are on the market typically only focus on one hazard at a time such as impact sensing, carbon monoxide detection, and health monitoring. SmartHat provides impact sensing, heart rate monitoring, carbon monoxide detection, UV monitoring, automatic headlamp, and real time cellular alerts into one integrated system. SmartHat combines a complete workplace hazard monitoring solution with cost effective implementation, making it the optimal choice for modern industrial safety environments.

4. Market Summary

4.1 Market Analysis

The construction and industrial safety market continues to expand as regulations and workplace safety enforcement become increasingly stringent. Companies in manufacturing, utilities, oil and gas, and large-scale construction projects must monitor environmental hazards, including carbon monoxide exposure, to remain compliant with OSHA standards. SmartHat directly addresses these compliance requirements, making it an attractive solution for employers seeking to protect their workforce, reduce liability, and improve operational efficiency. Beyond construction, the product has potential applications in any industrial setting where worker safety is a priority, including refineries, power plants, and chemical facilities, allowing for entry into multiple segments of the broader industrial safety market.

4.2 Marketing Strategy

Marketing for SmartHat will focus on direct engagement with contractor safety managers through targeted outreach, OSHA-aligned presentations, and hands-on demonstrations or pilot programs that showcase the product's real-world value. Participation in industry safety conferences will further highlight SmartHat's capabilities and position it as a trusted safety solution. A strong digital presence will complement these efforts, reinforcing credibility, increasing awareness, and providing easy access to product information for potential buyers.

4.3 Sales Projection

The SmartHat is projected to retail at $950 per unit, with a manufacturing cost of $600, resulting in a gross profit of $350 per unit. In the first year, sales are expected to reach 100 units, generating $95,000 in revenue and $35,000 in gross profit. By the second year, as contractor partnerships grow and regional expansion begins, unit sales are projected to increase to 250, yielding $237,500 in revenue and $87,500 in gross profit. By the third year, sales could reach 500 units, translating to $475,000 in revenue and $175,000 in gross profit. This growth reflects the scaling of partnerships, increased market awareness, and adoption across multiple industrial sectors.

4.4 Manufacturing and Distribution Strategy

SmartHat manufacturing will begin with small-batch assembly, using components sourced from established and reliable suppliers. All assembly and quality testing will be performed in-house to ensure consistency, durability, and adherence to safety standards. Initially, distribution will focus on direct sales to companies, allowing for close customer engagement and feedback. As demand grows, the strategy may expand to include regional distributors, enabling broader market reach while maintaining product quality and service standards.

5. Organization Overview

5.1 Company Structure

SmartHat LLC is structured as a member-managed limited liability company. The founding members will collectively oversee operations, technical development, and marketing, leveraging their combined engineering expertise. Landon Moore, as Project Manager, will coordinate overall operations, manage timelines, and ensure effective team collaboration. Matthew Dykes, serving as Technical Leader, will guide hardware and software development and oversee system integration. Dawson Carnathan, Aaron Ramsay, and Leigh Kergosien will support engineering development, including design, prototyping, and testing, contributing to both hardware and embedded software tasks. Together, the team will make strategic decisions, ensure quality and compliance, and drive SmartHat from concept through deployment.

5.2 Personnel Plan

During Year 1, the founding members of SmartHat LLC will manage all operations, development, and marketing responsibilities without taking a salary, investing their time and expertise to establish the company and product. As production volume increases in Years 2 and 3, additional personnel, such as assembly technicians, may be hired to support manufacturing and quality testing. The team maintains a collaborative approach, with members flexible to assist across roles, and will hold open discussions regarding company direction, product development, and financial management to ensure the business grows efficiently and strategically.

6. Financial Plan

The financial plan for SmartHat LLC projects operations over the first three years, with costs and revenues estimated based on market research, component pricing, and expected adoption within the industrial safety sector. Sales projections reflect a conservative approach to market entry, assuming initial direct sales to contractors and regional expansion over time. Operating expenses were estimated based on the needs of marketing, insurance, shipping, and administrative support, while production costs were calculated from component prices, assembly labor, and in-house quality testing.

6.1 Profit and Loss

The sales for SmartHat are generated by selling the product directly to industrial contractors and companies. Table 6.1 illustrates the predicted sales and costs to operate the SmartHat business for the first three years of operation.

Table 6.1 Profit & Loss

Income	Startup	Year 1	Year 2	Year 3
Units Sold	-	$100.00	$250.00	$500.00
Sales	$5,000.00	$95,000.00	$237,500.00	$475,000.00
Cost of Goods Sold	$1,000.00	$60,000.00	$150,000.00	$300,000.00
Gross Profit	$4,000.00	$35,000.00	$87,500.00	$175,000.00
Gross Margin %	-	36.8%	36.8%	36.8%
Operating Expenses				
Travel	$0.00	$3,000.00	$3,000.00	$3,000.00
Advertising	$0.00	$3,500.00	$5,000.00	$7,000.00
Shipping	$0.00	$1,000.00	$2,000.00	$4,000.00
Salaries	$0.00	$0.00	$0.00	$10,000.00
Capital Expenditures	$2,000.00	$2,000.00	$2,500.00	$3,000.00
Miscellaneous	$0.00	$500.00	$750.00	$1,000.00
Total Operating Expenses	$2,000.00	$10,000.00	$14,250.00	$25,000.00
Operating Profit	$2,000.00	$25,000.00	$73,250.00	$150,000.00
Taxes (15%)	$0.00	$3,750.00	$10,988.00	$22,500.00
Net Profit After Tax	$2,000.00	$21,250.00	$62,262.00	$127,500.00
Net Margin %	-	22.4%	26.2%	26.8%

The estimated sales are modest in the first year, with growth each year as advertising and outreach expand to the target market. The estimated cost of producing one SmartHat unit is approximately $600, which includes components, assembly, and in-house testing. Each unit will be sold at a retail price of $950, offering a competitive solution that combines safety, health, and environmental monitoring in a single wearable product. The team anticipates selling 100 units in the first year, 250 units in the second year, and 500 units in the third year.

Operating expenses capture the ongoing costs of maintaining and running the business. Recurring salaries are initially zero, as founders manage operations without pay; salaries are projected to begin in Year 3 with the hiring of assembly technicians to support higher production volumes. Insurance, travel, advertising, capital expenditures, and shipping costs are based on market research and scaled according to projected sales growth. Rent, utilities, and building maintenance are minimal since assembly and operations will initially take place in university facilities or team-managed spaces.

This structure allows SmartHat LLC to achieve profitability early while reinvesting in product development and market expansion. The projected net profit increases each year as sales grow and fixed costs are leveraged, demonstrating a sustainable financial outlook for the first three years of operation.

6.2 Projected Cash Flow

Table 6.2 shows the projected cash flow from the SmartHat Product.

Category	Startup	Year 1	Year 2	Year 3
Sales	$5,000.00	$95,000.00	$237,500.00	$475,000.00
Total Inflow	$5,000.00	$95,000.00	$237,500.00	$475,000.00
Materials (COGS)	$1,000.00	$60,000.00	$150,000.00	$300,000.00
Travel	$0.00	$3,000.00	$3,000.00	$3,000.00
Advertising	$0.00	$3,500.00	$5,000.00	$7,000.00
Shipping	$0.00	$1,000.00	$2,000.00	$4,000.00
Salaries	$0.00	$0.00	$0.00	$10,000.00
Capital Expenditures	$2,000.00	$2,000.00	$2,500.00	$3,000.00
Miscellaneous	$0.00	$500.00	$750.00	$1,000.00
Total Outflow	$3,000.00	$70,000.00	$161,250.00	$325,000.00
Net Cash Flow	$2,000.00	$25,000.00	$76,250.00	$150,000.00
Beginning Cash	$0.00	$2,000.00	$27,000.00	$103,250.00
Ending Cash Balance	$2,000.00	$27,000.00	$103,250.00	$253,250.00

The cash flow chart predicts a linear increase in profit, since we will be selling more products each year for SmartHat. The startup will cost around $1000 for materials and development, which is budgeted by Mississippi State University for the Senior Design course.

6.3 Projected Balance Sheet

Table 6.3 Projected Balance Sheet

Assets	Startup	Year 1	Year 2	Year 3
Cash	$2,000.00	$27,000.00	$103,250.00	$253,250.00
Equipment (Cumulative CapEx)	$2,000.00	$4,000.00	$6,500.00	$9,500.00
Total Assets	$4,000.00	$31,000.00	$109,750.00	$262,750.00
Liabilities				
Loans	$0.00	$0.00	$0.00	$0.00
Accounts Payable	$0.00	$9000.00	$22,500.00	$45,000.00
Total Liabilities	$0.00	$9,000.00	$22,500.00	$45,000.00
Equity				
Owner Investment	$2,000.00	$2,000.00	$2,000.00	$2,000.00
Retained Earnings (Cumulative)	$2,000.00	$20,000.00	$85,250.00	$215,750.00
Total Equity	$4,000.00	$22,000.00	$87,250.00	$217,750.00

Table 6.3 presents the projected balance sheet for SmartHat over the first three years of operation. Total assets increase each year due to rising cash balances generated from positive net income and cumulative capital expenditures for equipment. Cash growth reflects strong operating performance and reinvestment of profits.

Accounts payable are projected at 15% of annual cost of goods sold to reflect a realistic short-term liability. No long-term debt is assumed during the first three years to minimize financial risk during early stages. Retained earnings increase annually as profits are reinvested into the company. The balance sheet shows: Assets = Liabilities + Equity**.**

* The authors acknowledge the use of ChatGPT in the preparation of this assignment for phrasing and proofreading.